Sensorless Control of Switched Reluctance Motor Based on Phase Inductance Vectors

Jun Cai and Zhiquan Deng

***Abstract*—A new sensorless control method based on phase inductance vectors of switched reluctance motor (SRM) is presented in this paper. To implement this method, the following was done. First, based on the pulse injection technique, a new method called phase current slope difference is proposed for identifying the full-cycle phase inductance of each phase. Second, the phase inductances are considered as vectors with fixed phase difference and the composed vector is rotated with the varied rotor position. Combining the composed vector orthogonal decomposition method and a new inductance subregional method, the rotor position can be estimated. This method can realize sensorless control of SRM from starting up at standstill to high-speed operation and is quite easy to implement. No additional hardware or complex computation is required and no *a priori* knowledge on magnetic characteristics of SRM is necessary. To verify the performance of the proposed sensorless scheme, the following experiments have been performed: 1) the initial rotor position at standstill and rotating shaft conditions are estimated for sensorless startup control; and 2) the rotor position and rotational speed are estimated at different reference speeds for sensorless speed closed-loop control and the dynamic performance is fully validated.**

***Index Terms*—Phase current slope difference, phase inductance subregional, phase inductance vector, sensorless control, switched reluctance motor (SRM).**

I. Introduction

SWITCHED reluctance motor (SRM) has received considerable attention in connection with variable speed motor drive applications due to its low cost, rugged structure, reliable converter topology, high efficiency over a wide speed range, and fault-tolerant performance. However, the basic operation of the SRM requires proper rotor position. Usually, mechanical position sensors, such as optical encoders and Hall sensors are used to provide the rotor position information for SRM drives, but they add to the cost and dimension and deteriorate the reliability of the SRM drives. Therefore, the development of a reliable and low-cost sensorless control method is quite necessary [1]–[3].

Several sensorless control methods have been reported in the literature over the past three decades [1]–[29]. The main idea behind these techniques stems from the fact that the electromagnetic characteristics of SRM, such as flux linkage, phase inductance, back EMF, etc., are the function of rotor position. These characteristics are usually estimated according to the phase voltage equation of SRM by terminal measurement of phase current and voltage, and then the rotor position information can be decoded accordingly. Various published methods use this principle differing only in the way the electromagnetic characteristics are measured and in the decoding methods used for rotor position information. As for the operating speed range, the estimation method can be classified into two categories. In the first category, high-frequency signals are introduced during the time when the phase is nonenergized to determine the unsaturated phase inductance characteristics, and from which the rotor position or phase excitation logics can be extracted [4]–[12], [29]. The second one utilizes the terminal measurement of phase voltage and current of the energized phase as inputs to obtain the characteristics such as flux linkage and phase inductance for rotor position estimation [14]–[29]. Generally, the former are suitable for low-speed operation, and the latter are more suitable for high-speed operation. But from the aspect of rotor position decoding methods, the following methods such as lookup table based [14]–[18], mathematical model based [19]–[21], intelligent approximation algorithm based [22]–[27], observer based methods [28], [29], etc., are included. This implies that to implement these estimation methods, some *a priori* knowledge of the SRM has to be known.

In the last two years, some new position estimation schemes have been reported. In [13], a method to estimate the initial position of SRM using bootstrap circuits is presented. This method has some advantages such as independence of any inductance parameter, requirement of lower bandwidth of control system, and reduced electromagnetic interference influence. However, the measurement of the charging time of bootstrap circuits is relatively cumbersome. In [30] and [31], a position estimation method by means of voltage measurements is proposed. The authors define a resonant circuit by combination of the motor and the power converter; the associated resonance frequency of the circuit depends on the rotor position. This method has some favorable characteristics such as measurement of large-amplitude voltages, robustness against temperature deviations, and high rates for position estimation. In [32], a low-speed sensorless strategy is discussed. By adapting the test signals to the steady-state stator voltage, the distortion of the test pulses can be reduced. In addition, some new applications of sensorless methods such as coordinate transformation, high-frequency injection, fuzzy logics, and MARS observers in other kind of electric machine are discussed in [33]–[36], respectively, which

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can be a good guide for developing new sensorless schemes for SRM drives.

For no-load and light-load applications, the amplitude of the resulting current is relatively small; thus, the measured inductance is effectively the unsaturated inductance. In this application, the diagnose signal-injection-based methods are simple and practical. The main principle of this method is that the phase inductance varies with rotor positions periodically, by injecting high-frequency signals into the nonenergized phase and the resulting current will change with rotor position. In [4] and [5], modulation techniques are used for decoding the rotor position. In this method, sinusoidal voltage is applied in the idle phase; thus, the amplitude and phase of the result current contain encoded rotor position information. But this method needs external circuitry for signal injection and rotor position demodulation. In [29], high-frequency voltage pulses are provided by existing power converter. The frequency of injected pulse is chosen such that the amplitude of the resulting current remains small and does not produce negative torque. The shape of the current pulse is a function of phase inductance. Accordingly, the rotor position can be estimated from the mapping of inductance to position. But this method requires some *a priori* knowledge of phase inductance characteristics. Besides, the angular speed is assumed to be very small and the back EMF is neglected. Therefore, the application speed range of this method is restricted seriously. To solve this problem, in [8], an improved impressed pulse technique is proposed. This method only requires injecting pulses into one nonenergized phase; by using a low-pass filter, the pulse currents can be turned into an approximate sinusoidal waveform, the peak of which and the waveform of inductance variation are in inverse. By availing of the filter lagging angle, the reference value of threshold can be estimated, which is corresponding to the starting points of exciting phases, respectively. This method simplifies the control algorithm and facilitates the extension in speed control range. However, the satisfactory performance of the low-pass filter cannot be obtained if the rotational speed is lower than a certain value. Therefore, this method still needs other sensorless methods for starting up.

As discussed in past literature [9]–[14], the sensorless starting-up technique is one of the key research problems in sensorless control of SRM. In [14], the diagnostic signal and a comprehensive set of magnetic data are used to detect the rotor position at standstill. In [9], each phase is excited in sequence by short voltage pulses, and the amplitude order of phase inductance can be detected from the order of phase current. Therefore, the phase capable of producing positive torque can be chosen for driving the SRM. This method is easy to implement without any computation, but the initial position is not estimated. In [10], all phases are excited for short duration for choosing the optimal phase current to estimate rotor position by cubic spline technique and identify the phase in positive torque region at standstill. This method can estimate the initial angle at standstill, but the spline function of the rotor position and phase current is not easy to model. In [11], based on the pulse injection method, the phase inductance of each phase can be estimated from voltage equation, which has the same waveform and different phase angle. Therefore, the inductance can be represented as vectors with fixed phase angle difference. Using the vector orthogonal decomposition method, the initial electrical angle at standstill can be estimated by solving a tangent function. But this paper does not take the definitional domain of tangent function or the range of the antitangent function $(-\pi/2, \pi/2)$ into account, thereby how to deal with the function range to get the full-cycle rotor position information is still a problem. Moreover, the paper does not discuss the application feasibility of this method for sensorless control at different operation conditions. In [12], a method for the sensorless start up from standstill and rotating shaft conditions has been introduced, which uses the time required for reaching a judiciously selected threshold in the diagnostic current for position estimation. This method eliminates the problem of starting hesitation and is easy to implement. But the position estimation performance of this method for low-speed to high-speed operation is not discussed.

In this paper, we present a new sensorless control scheme for SRM drives, which has the following primary contributions: 1) this method can estimate the initial rotor position at standstill and those with rotating shaft conditions accurately and start up the SRM drives without starting hesitation; 2) this method can drive the SRM from standstill to high-speed operation and the rotor position can be estimated in real time; 3) the estimation performance in inductance asymmetric conditions is analyzed quantitatively; 4) all those traditional SRM closed-loop control techniques can be implemented easily in this sensorless control mode; and 5) this method does not require any additional hardware and occupy a huge amount of memory; in addition, no complex computation process and no *a priori* knowledge on magnetic characteristics of SRM are required.

To implement this method, two steps are necessary: 1) a new identification method for full-cycle phase inductance of SRM is proposed; and 2) a new full-cycle phase inductance subregional method combined with orthogonal decomposition of the three-phase inductance composed vector is proposed for rotor position estimation. Theoretical analysis and experiments are included in the following sections to validate the proposed sensorless method.

This paper is organized as follows. In Section II, the theoretical analysis of the identification method for full-cycle phase inductance is discussed. The principle of the proposed position estimation methods is discussed in Section III. Section IV briefly discusses the experimental results of the sensorless control method. Section V concludes the paper.

II. Full-Cycle Inductance Identification Technique

A. Fundamentals of the Full-Cycle Inductance Identification Method

In this paper, a three-phase 12/8 structure SRM prototype is used as the study object and the power converter is the classical three-phase asymmetric half bridge circuits. The configuration of the power converter is shown in Fig. 1(a).

As is known, the SRM is sequentially energized to produce suitable positive torque for motoring operation. Therefore, in order to obtain the full-cycle self inductance of SRM, both of the energized phase and unenergized phase should be considered. In

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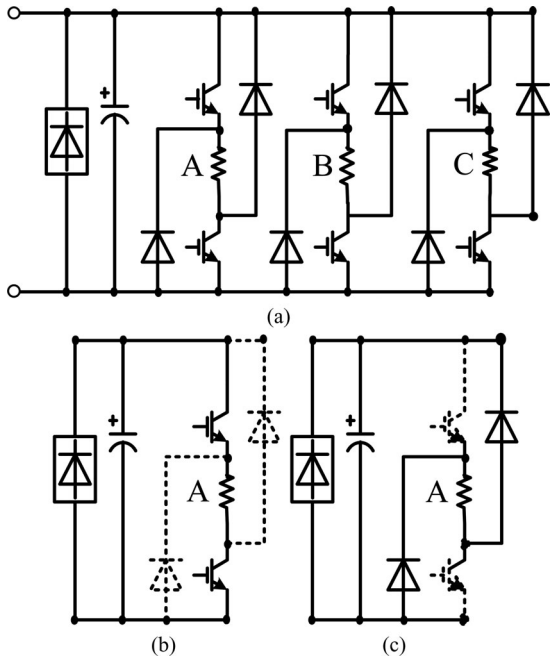


Fig. 1. (a) Three-phase asymmetric half bridge power converter. (b) Turn-ON mode. (c) Turn-OFF mode.

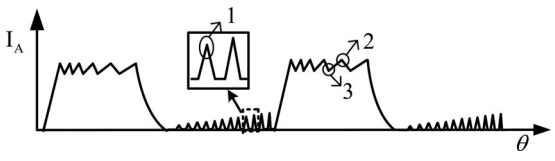


Fig. 2. Typical phase current waveform.

this section, a phase current slope difference method is proposed. Here, we let the converter work in hard switching modes, where the upper and lower insulated gate bipolar transistors in the same phase leg are both turn-ON and turn-OFF. The different equivalent circuits corresponding to the two switching modes applied to phase-A are shown in Fig. 1(b) and (c), respectively. As a result, the phase voltage changes between $U_{\rm dc}$ and $-U_{\rm dc}$, where $U_{\rm dc}$ is the bus voltage.

In nonenergized phases, series of high-frequency voltage pulses provided by existing power converter are injected. While in energized phases, no voltage pulses are injected. The SRM is working in current hysteresis chopping control (CCC) mode and the phase current is controlled between two current levels equal to $i_{\rm chop} \pm i_{\rm band}$, where $i_{\rm chop}$ is the reference chopping current and $i_{\rm band}$ is the hysteresis band. Therefore, the power converter works in turn-ON and turn-OFF mode commutatively and continuously. The typical phase current waveform is shown in Fig. 2.

For no-load or light-load application modes, the magnetic saturation can be neglected. Thus, if the voltage drop in power switches and diodes is neglected, when the converter is working in turn-ON mode, the phase voltage equation can be expressed as

$$U_{\rm dc} = Ri + L(\theta)\left.\frac{di}{dt}\right|_{\rm on} + i\omega\frac{dL(\theta)}{d\theta}. \tag{1}$$

While in turn-OFF mode, the phase voltage equation can be expressed as

$$-U_{\rm dc} = Ri + L(\theta)\left.\frac{di}{dt}\right|_{\rm off} + i\omega\frac{dL(\theta)}{d\theta}. \tag{2}$$

Here, $U_{\rm dc}$, R, i are the bus voltage, phase resistance, and phase current, respectively; $L(\theta)$ is the self-inductance; ω is the angular speed, and $di/dt|_{\rm on}$ and $di/dt|_{\rm off}$ are the positive and negative slopes of a given phase current, respectively.

For the switching time interval between turn-ON mode and turn-OFF mode is short enough, it is assumed that the current in these two switching modes is almost constant. Therefore, the voltage drops of the resistance in these two switching modes are approximately equal and the back EMFs in these two modes are also similar. Then, the phase inductance can be derived as

$$L(\theta) = \frac{2U_{\rm dc}}{di/dt|_{\rm on} - di/dt|_{\rm off}} = \frac{2U_{\rm dc}}{\Delta I} \tag{3}$$

where $\Delta I = di/dt|_{\rm on} - di/dt|_{\rm off}$ is the current slope difference.

From (3), assuming the voltage applied is known, the phase inductance is in inverse proportion to the difference of positive and negative slopes of phase current. This method eliminates the effects of back EMF indirectly, which can enhance the estimation precision and extend the applicable speed range effectively.

As shown in Fig. 2, in region 1 and region 2, the phase current slope difference between the turn-ON and turn-OFF interval can be measured. Similarly, in region 3, the phase current slope difference from turn-OFF to turn-ON interval can also be measured. Therefore, a series of discrete numerical differentiation of phase current can be obtained, and using the maximum filtering method, the envelope curve of which can be filtered out. Then, the phase inductance can be estimated from (3). The flow chart and principle diagram of full-cycle phase inductance estimation is shown in Figs. 3 and 4(a). As shown in Fig. 4(a), for a three-phase 12/8 structure SRM prototype, the rotor position θ is defined as $\theta = 0^\circ$ at the unaligned position and $\theta = 22.5^\circ$ at the aligned position. The phase inductance varies with the rotor position periodically.

B. Estimation of Inductance at Standstill and With Rotating Shaft Conditions

The inductance identification method discussed earlier can also be applied to estimate the inductance at standstill and those with rotating shaft conditions.

At standstill, all phases are injected with high-frequency voltage pulses. To keep the rotor from moving, suitable pulse frequency should be chosen. When the applied voltage is $U_{\rm dc}$, the phase inductance can be estimated by (3). Based on the estimation principle flow chart as shown in Fig. 2, the phase inductance can be measured. Fig. 4(b) shows the realization principle diagram.

The rotating shaft conditions are defined as follows: 1) all phases are clear and no remaining currents; 2) no driving signals are given, and the rotor is in free-running mode due to its own residual kinetic energy. In order to estimate the phase

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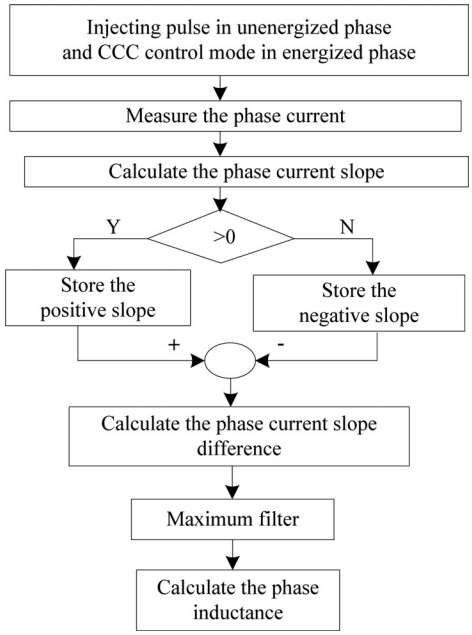


Fig. 3. Flow chart of full-cycle phase inductance estimation.

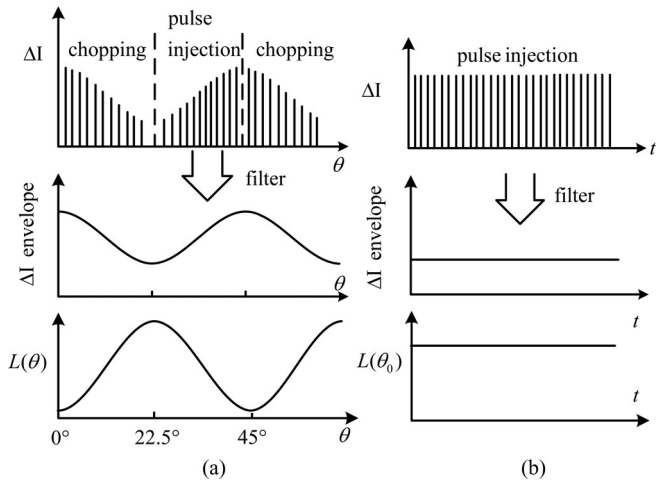


Fig. 4. Principle diagram of full-cycle inductance estimation.

inductance in this condition, the chopping regions, as shown in Fig. 4(a), are substituted by injecting high-frequency pulse signals. When all three phases are injected with short pulses, each phase inductance can also be estimated by the phase current slope difference-based method.

C. Selection of Parameters of Injected Pulses

1) Selection of the Maximum Test Excitation Duration: Since the SRM is not saturated during test excitation, the electric torque T_e can be represented by

$$T_e = \frac{1}{2} i^2 \frac{dL(\theta)}{d\theta} \tag{4}$$

where i is the phase current and $dL(\theta)/d\theta$ is the inductance slope, which can be represented by (5) and (6), respectively.

$$i = \frac{U_{\rm dc}\Delta t}{L} \le \frac{U_{\rm dc}\Delta t}{L_{\min}} \tag{5}$$

$$\frac{dL(\theta)}{d\theta} = \frac{L_{\max} - L_{\min}}{\beta_s} \tag{6}$$

where $U_{\rm dc}$ is the bus voltage, Δt is the excitation duration, $L_{\max}$ and $L_{\min}$ are the maximum and minimum inductance, respectively, and β_s is the stator pole arc.

Supposing the overall friction torque on the shaft is T_f. To keep the rotor from moving, it should be satisfied that

$$T_e < T_f. \tag{7}$$

Combining (4)–(7), the maximum excitation duration $\Delta t_{\max}$ can be derived as follows:

$$\Delta t_{\max} < \frac{L_{\min}}{U_{\rm dc}} \sqrt{2T_f \frac{\beta_s}{L_{\max} - L_{\min}}}. \tag{8}$$

2) Selection of the Minimum Test Excitation Duration: As is known, reducing the excitation period can lower the power loss and reduce the negative torque. However, as the current peak becomes lower, the measuring resolution and the capability of antiinterference are reduced. Therefore, the pulse current should be significant enough for the current sampling system. Supposing the minimum phase current that can be measured is $i_{\min}$, the minimum test excitation duration $\Delta t_{\min}$ can be estimated as

$$\Delta t_{\min} = \frac{L_{\max} \cdot i_{\min}}{U_{\rm dc}}. \tag{9}$$

In the tested SRM, $U_{\rm dc} = 20$ V, $L_{\max} = 1.506$ mH, $L_{\min} = 0.344$ mH, $i_{\min} = 0.2$ A, $T_f = 0.05$ N·m, and $\beta_s = 15.858°$. We can obtain that $\Delta t_{\max} = 635.4$ μs and $\Delta t_{\min} = 15.06$ μs. In this paper, the pulse excitation duration Δt is chosen to be 100 μs. When the duty ratio of the pulse is set to 1/3, the pulse current can decay to zero before the next pulse injection starts. Thus, the pulse frequency is 3.33 kHz.

III. Proposed Sensorless Control Method

A. Fundamentals of the Phase Inductance Vector-Based Rotor Position Estimation Method

Based on the estimation method as discussed earlier, the three-phase current slope difference and the full-cycle inductance profiles of the tested 12/8 SRM for an electrical cycle are shown in Fig. 5, which vary with the rotor position periodically. In this paper, the rotor angular position θ is defined as $\theta = 0°$ at the unaligned position and $\theta = 22.5°$ at the aligned position. Without considering the electrical asymmetric of the SRM prototype, the waveforms of each phase inductance are the same, but with a fixed phase difference that is equal to one step angle 15° among them. For a 12/8 SRM, a mechanical cycle is 45°, by (10), the mechanical angle θ can be converted to electrical angle $\theta_{\rm elec}$.

$$\theta_{\rm elec} = N_r \theta \tag{10}$$

where $N_r = 8$ is the number of rotor poles.

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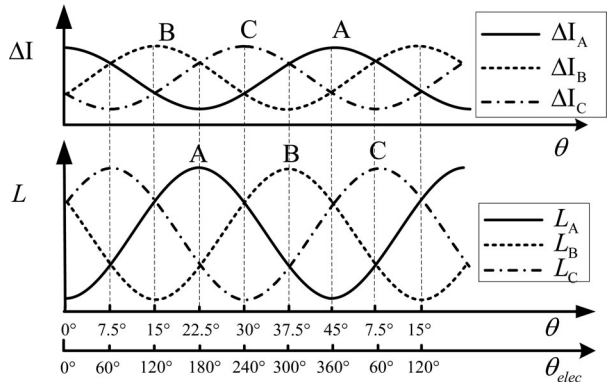


Fig. 5. Waveforms of three-phase current slope difference and full-cycle inductance.

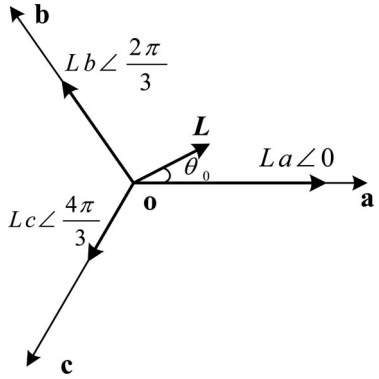


Fig. 6. Three-phase inductance vectors.

$\theta_{\rm arc}$ is the radian representation of the electrical angle $\theta_{\rm elec}$, which can be represented by

$$\theta_{\rm arc} = \frac{\pi}{180}\theta_{\rm elec}. \tag{11}$$

Due to these characteristics, the three-phase inductance can be represented by three positive vectors with fixed phase difference of 120° electrical angle, and the norm of which is the magnitude of the phase inductance. As shown in Fig. 6, the three-phase inductance vectors ***La***, ***Lb,*** and ***Lc*** are relatively static in the **a-b-c** plane coordinate system, where $La\angle 0$, $Lb\angle 2\pi/3$, and $Lc\angle 4\pi/3$ represent the relative positions of each phase inductance vector, respectively, and the composed vector $\boldsymbol{L}$ can be expressed by

$$L = La\angle 0 + Lb\angle\frac{2\pi}{3} + Lc\angle\frac{4\pi}{3}. \tag{12}$$

For the vectors ***La, Lb,*** and ***Lc*** are fixed in **a, b,** and **c** coordinated axis of the **a-b-c** coordinate system, with the change of rotor position, the amplitude of each phase inductance will change periodically and the composed inductance vector $\boldsymbol{L}$ will rotate with the same angular speed of SRM relative to the **a-b-c** coordinate system. Therefore, the rotor position can be decoded by orthogonal decomposition of $\boldsymbol{L}$**.** As shown in Fig. 6, when the inductance amplitude of one phase reaches its maximum value, the composed inductance vector $\boldsymbol{L}$ will superposition completely with this phase inductance vector. For example, when $\theta_{\rm arc} = \pi$ and La reaches its maximum value, then $\boldsymbol{L}$ superposition with ***La*** and $\theta_0 = 0$; therefore, the relationship between $\theta_{\rm arc}$ and θ_0 can be expressed by

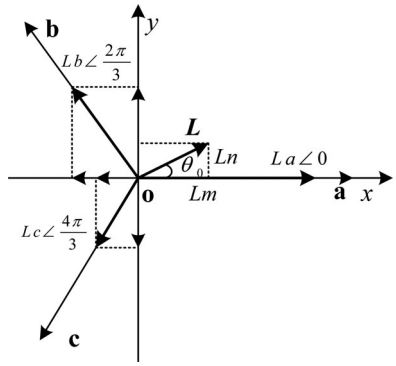


Fig. 7. Diagram of orthogonal decomposition of ***L.***

$$\theta_{\rm arc} = \pi + \theta_0. \tag{13}$$

Combining (10), (11), and (13), the mechanical rotor position θ can be obtained as

$$\theta = \frac{180}{N_r\pi}[\pi + \theta_0]. \tag{14}$$

As shown in Fig. 7, by using orthogonal decomposition method, the x-axis component Lm and the y-axis component Ln can be represented by

$$Lm = La - \frac{1}{2}(Lb + Lc) \tag{15}$$

$$Ln = \frac{\sqrt{3}}{2}(Lb - Lc). \tag{16}$$

And the phase angle of $\boldsymbol{L}$ can be obtained from

$$\theta_0 = \arctan\left(\frac{Ln}{Lm}\right). \tag{17}$$

Because the domain of Arctan function is $(-\pi/2, \pi/2)$, combining (14) and (17), it can be obtained that $\theta_0 \in(-\pi/2, \pi/2)$ and $\theta \in(11.25°, 33.75°)$. In order to estimate the full cycle rotor position, a new phase inductance subregional method is adopted in this paper.

B. Fundamentals of the New Phase Inductance Subregional Method

Compare the amplitudes of three-phase inductance and choose the phase with the largest inductance as the norm estimation phase. In other words, by detecting the order of inductance magnitudes among all three phases, one can uniquely detect a region with a length of 120° electrical angle. This in turn will enable zooming into the targeted phase region for precise calculation of the rotor position rather than considering the calculation problem caused by the domain of the Arctan function as discussed earlier. Combining the estimated rotor position in each region, the rotor positions in the full electrical cycle can be obtained. Then, the mechanical rotor position can be calculated from (14). The subregional logics of the sensorless method are given in Table I. As can be

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TABLE I
SUBREGIONAL LOGICS OF THE SENSORLESS METHOD

Amplitudes order of phase inductance	Position Region	Norm estimation phase
$La \geqslant Lc$ & $La > Lb$	Region 1 $\theta_{arc} \in [2\pi/3, 4\pi/3)$	A
$Lb \geqslant La$ & $Lb > Lc$	Region 2 $\theta_{arc} \in [4\pi/3, 2\pi)$	B
$Lc \geqslant Lb$ & $Lc > La$	Region 3 $\theta_{arc} \in [0, 2\pi/3)$	C
return	…	…

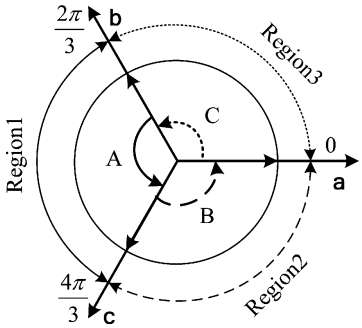


Fig. 8. Position estimation regions using the new subregion method.

seen in Table I, the phase with the largest inductance is chosen as the norm estimation phase, and accordingly the three-phase full-cycle inductance of an electrical cycle is divided into three different regions, such as $\theta_{\rm arc} \in [2\pi/3, 4\pi/3)$, $\theta_{\rm arc} \in [4\pi/3, 2\pi)$, and $\theta_{\rm arc} \in [0, 2\pi/3)$. The inductance subregion method is illustrated in Fig. 8. Based on the subregional method, the rotor position θ can be estimated in each position region, respectively, which is discussed in detail in the following section.

C. Fundamentals of the New Rotor Position Estimation Method

Based on the inductance subregional method, the rotor position can be estimated in each respective region. In order to analyze the calculation process more clearly, the **a-b-c** coordinate system is treated correspondingly. When the norm estimation phase is chosen, the **a-b-c** coordinate system is rotated with corresponding phase electrical angle to let the inductance vector of the norm estimation phase superposition with the x axis. The position estimation processes are as follows.

1) When $La \geq Lc$ and $La > Lb$, the rotor position is located in Region 1, where $\theta_{\rm arc} \in [2\pi/3, 4\pi/3)$, $\theta_0 \in [-\pi/3, \pi/3)$.

As shown in Fig. 9, after orthogonal decomposition of $\boldsymbol{L}$, the x-axis component *Lma* and the y-axis component *Lna* can be calculated by (15) and (16), and the phase angle of $\boldsymbol{L}$ can be obtained by

$$\theta_a = \arctan\left(\frac{Lna}{Lma}\right) \theta_a \in \left[-\frac{\pi}{3}, \frac{\pi}{3}\right) \tag{18}$$

$$\theta_0 = \theta_a. \tag{19}$$

Therefore, the mechanical rotor position θ can be obtained from (14).

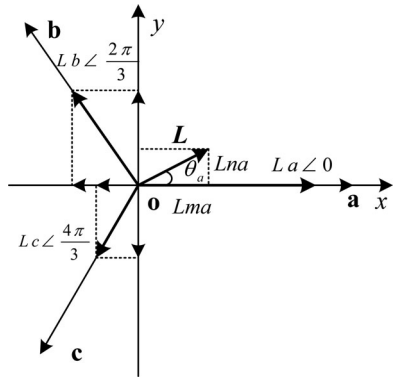


Fig. 9. Diagram of orthogonal decomposition of $\boldsymbol{L}$ in Region 1.

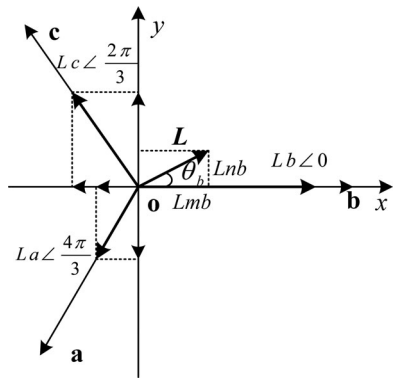


Fig. 10. Diagram of orthogonal decomposition of $\boldsymbol{L}$ in Region 2.

2) When $Lb \geq La$ and $Lb > Lc$, the rotor position is located in Region 2, where $\theta_{\rm arc} \in [4\pi/3, 2\pi)$, $\theta_0 \in [\pi/3, \pi)$.

In this region, the ***xy*** orthogonal coordinate is fixed, but the **a-b-c** coordinate system has rotated $2\pi/3$ rad in clockwise. After coordinate transformation, the composed inductance vector $\boldsymbol{L}$ can be represented by

$$\boldsymbol{L} = Lb\angle 0 + Lc\angle\frac{2\pi}{3} + La\angle\frac{4\pi}{3}. \tag{20}$$

The orthogonal decomposition result of the composed inductance vector $\boldsymbol{L}$ is shown in Fig. 10.

As shown in Fig. 10, the x-axis component *Lmb* is given by

$$Lmb = Lb - \frac{1}{2}(La + Lc) \tag{21}$$

and the y-axis component *Lnb* is represented by

$$Lnb = \frac{\sqrt{3}}{2}(Lc - La). \tag{22}$$

Then, the phase angle of $\boldsymbol{L}$ can be calculated from

$$\theta_b = \arctan\left(\frac{Lnb}{Lmb}\right), \quad \theta_b \in \left[-\frac{\pi}{3}, \frac{\pi}{3}\right). \tag{23}$$

Therefore, the corresponding phase angle of $\boldsymbol{L}$ before coordinate transformation is

$$\theta_0 = \theta_b + \frac{2\pi}{3}. \tag{24}$$

As with the case in region 1, θ can also be obtained from (14).

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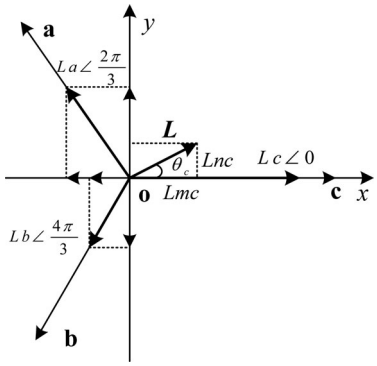


Fig. 11. Diagram of orthogonal decomposition of $\boldsymbol{L}$ in Region 3.

3) When $Lc \geq Lb$ and $Lc > La$, the rotor position is located in Region 3, where $\theta_{\rm arc} \in [0, 2\pi/3)$, $\theta_0 \in [-\pi, -\pi/3)$.

In this region, the **a-b-c** coordinate system is rotated $4\pi/3$ rad in clockwise, Fig. 11 shows the results. After coordinate transformation, the composed inductance vector $\boldsymbol{L}$ can be represented by

$$\boldsymbol{L} = Lc\angle 0 + La\angle\frac{2\pi}{3} + Lb\angle\frac{4\pi}{3}. \tag{25}$$

As shown in Fig. 11, the x-axis component *Lmc* and the y-axis component *Lnc* are given by

$$Lmc = Lc - \frac{1}{2}(Lb + La) \tag{26}$$

$$Lnc = \frac{\sqrt{3}}{2}(La - Lb). \tag{27}$$

Then, the phase angle of $\boldsymbol{L}$ can be obtained

$$\theta_c = \arctan\left(\frac{Lnc}{Lmc}\right), \quad \theta_c \in \left[-\frac{\pi}{3}, \frac{\pi}{3}\right). \tag{28}$$

And the corresponding phase angle of $\boldsymbol{L}$ before coordinate transformation is obtained

$$\theta_0 = \theta_c - \frac{2\pi}{3}. \tag{29}$$

Therefore, θ can be obtained from (14).

Combining (14), (19), (24), and (29), the estimated mechanical rotor position can be represented by

$$\theta = \begin{cases} \dfrac{22.5}{\pi}(\pi + \theta_a) & \text{(in Region 1)} \\ \dfrac{22.5}{\pi}\left(\theta_b + \dfrac{5\pi}{3}\right) & \text{(in Region 2)} \\ \dfrac{22.5}{\pi}\left(\theta_c + \dfrac{\pi}{3}\right) & \text{(in Region 3).} \end{cases} \tag{30}$$

And then, the rotational speed can be estimated by

$$\omega(n) = \frac{(\theta(n) - \theta(n-1)) \cdot f_{t1}}{6} \tag{31}$$

where $\theta(n)$ and $\theta(n-1)$ are the sampled rotor position in the neighboring two sample cycle, f_{t1} is the interrupt frequency of DSP timer 1 and $\omega(n)$ is the rotational speed (the unit is "r/min").

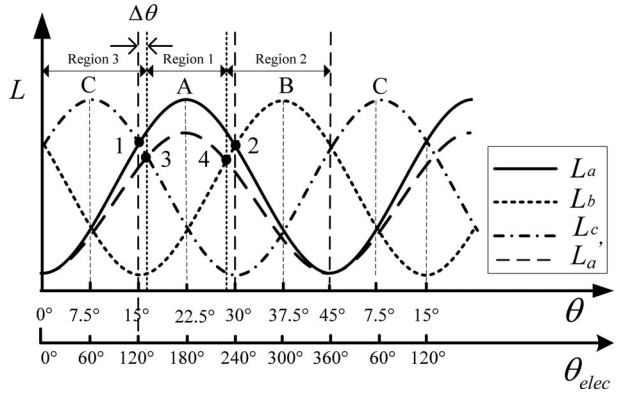


Fig. 12. Three-phase inductance with electrical asymmetric.

TABLE II
NEW SUBREGIONAL LOGICS OF THE SENSORLESS METHOD

Amplitudes order of phase inductance	Position Region	Norm estimation phase
La≥*Lc* & *La*>*Lb*	Region 1 $\theta_{arc} \in [2\pi/3 + \Delta\theta, 4\pi/3 - \Delta\theta)$	A
Lb≥*La* & *Lb*>*Lc*	Region 2 $\theta_{arc} \in [4\pi/3 - \Delta\theta, 2\pi)$	B
Lc≥*Lb* & *Lc*>*La*	Region 3 $\theta_{arc} \in [0, 2\pi/3 + \Delta\theta)$	C
return	…	…

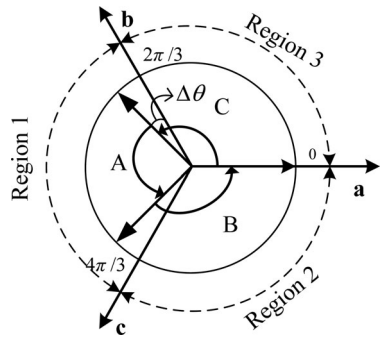


Fig. 13. Position estimation regions.

D. Performance Analysis of the Rotor Position Estimation Method Under Inductance Asymmetric Conditions

In the previous sections, the rotor position estimation method is analyzed based on the premise that each phase inductance is strictly symmetrical. As is known, ideally each phase inductance of the SRM prototype is symmetric. However, sometimes due to the machining or measure errors, the phase inductance are not always strictly symmetrical. From this perspective, it is necessary to study the estimation performance of the sensorless method under the circumstances of asymmetric phase inductance.

As shown in Fig. 12, we assume that the A-phase inductance is slightly different from that in the other two phases. According to the inductance subregion principle discussed in Section III-B, the new subregional logics are given in Table II and illustrated in Fig. 13.

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In order to verify the performance of the estimation method in inductance asymmetric condition, the rotor position estimation error is analyzed quantitatively. Here, taking Region 3 as an example, the estimation error and its range can be calculated as follows.

At first, in different rotor positions, use (32) to represent the degree of asymmetric of inductance

$$\lim_{k\to 1} La' = \lim_{k\to 1} kLa = La \quad (k \to 1) \tag{32}$$

where k is the asymmetric coefficient, $k \in (0,1) \cup (1,+\infty)$. Since the analysis is similar in $k \in (0,1)$ and $k \in (1,+\infty)$; here, we take $k \in (0,1)$ for an example.

In Region 3, when (32) is satisfied, it can be proved by contradiction that

$$Lmc = Lc - \frac{1}{2}(Lb + La) > 0. \tag{33}$$

As a result, (28) is satisfied. Decomposing the inductance composed vector, the x-axis component *Lm* and the y-axis component *Ln* can be represented by

$$\begin{cases} Lmc' = Lc - \dfrac{1}{2}(Lb + La') \\ Lnc' = \dfrac{\sqrt{3}}{2}(La' - Lb). \end{cases} \tag{34}$$

The phase angle of $\boldsymbol{L}$ can be obtained

$$\theta_c' = \arctan\left(\frac{Lnc'}{Lmc'}\right). \tag{35}$$

Thus, the estimation error can be expressed by

$$\delta\theta_c = \theta_c - \theta_c' = \arctan\left(\frac{Lnc}{Lmc}\right) - \arctan\left(\frac{Lnc'}{Lmc'}\right). \tag{36}$$

Then

$$\tan(\delta\theta_c) = \frac{LncLmc' - Lnc'Lmc}{LmcLmc' + LncLnc'}. \tag{37}$$

Combining (26), (27), (34), and (37)

$$\delta\theta_c = \arctan\left(\frac{(\sqrt{3}/2)(Lc - Lb)(La - La')}{K_1}\right). \tag{38}$$

Substituting (32) into (38), we can obtain

$$\delta\theta_c = \begin{cases} 0 & (k = 1) \\ \arctan\left[\dfrac{(\sqrt{3}/2)La(Lc - Lb)}{(K_2/1 - k) - K_3}\right] & (k \in (0,1) \text{ and } k \to 1). \end{cases} \tag{39}$$

It can be proved easily that

$$K_2 \geq 0 \tag{40}$$

$$Lc - Lb \geq 0. \tag{41}$$

Therefore, the estimation error range can be obtained

$$\delta\theta_c \in \left[0, \arctan\left[\frac{(\sqrt{3}/2)La(Lc - Lb)}{(K_2/1 - k) - K_3}\right]\right] \quad (k \in (0,1) \text{ and } k \to 1) \tag{42}$$

where it is obvious that

$$\lim_{k\to 1} \arctan\left[\frac{(\sqrt{3}/2)La(Lc - Lb)}{(K_2/1 - k) - K_3}\right] = 0 \tag{43}$$

Thus, the position estimation error in Region 3 is approaching to zero when k is infinitely approaching to 1.

In the same way, the position estimation error and its range in the other two regions can also be obtained.

1) In Region 1:

The estimation error can be expressed by

$$\delta\theta_a = \arctan\left(\frac{(\sqrt{3}/2)(Lb - Lc)(La' - La)}{K_4 + (3/4)(Lb - Lc)^2}\right). \tag{44}$$

Substituting (32) into (44), we can obtain

$$\delta\theta_a = \begin{cases} 0 & (k = 1) \\ \arctan\left(\dfrac{(\sqrt{3}/2)(Lc - Lb)La}{K_5 + (K_6/1 - k)}\right) & (k \in (0,1) \text{ and } k \to 1). \end{cases} \tag{45}$$

And then, the estimation error range is

$$\begin{cases} \delta\theta_a = 0 & (k = 1) \\ \delta\theta_a \in \left(0, \arctan\left(\dfrac{(\sqrt{3}/2)(Lc - Lb)La}{K_5 + (K_6/1 - k)}\right)\right] & (Lc \geq Lb) \text{ and } (k \to 1) \\ \delta\theta_a \in \left[\arctan\left(\dfrac{(\sqrt{3}/2)(Lc - Lb)La}{K_5 + (K_6/1 - k)}\right), 0\right) & (Lc < Lb) \text{ and } (k \to 1). \end{cases} \tag{46}$$

2) In Region 2:

The estimation error can be expressed by

$$\delta\theta_b = \arctan\left(\frac{(\sqrt{3}/2)(Lb - Lc)(La' - La)}{K_1}\right). \tag{47}$$

Combining (32) and (47), the estimation error can be represented by

$$\delta\theta_b = \begin{cases} 0 & (k = 1) \\ \arctan\left(\dfrac{(\sqrt{3}/2)La(Lc - Lb)}{(K_2/1 - k) - K_3}\right) & (k \in (0,1) \text{ and } k \to 1). \end{cases} \tag{48}$$

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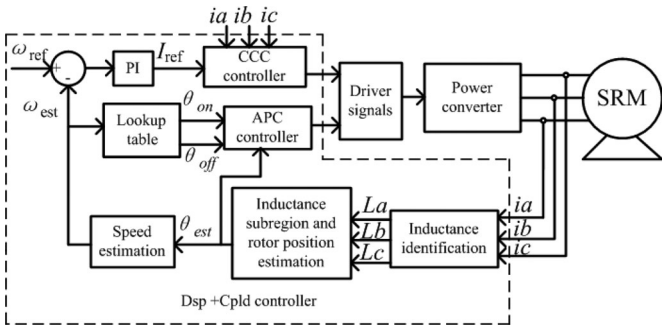


Fig. 14. Block diagram of the sensorless SRM drive.

In Region 2, $Lc \le Lb$; thus

$$\begin{cases} \delta\theta_b = 0 & (k=1) \\ \delta\theta_b \in \left[\arctan\left(\dfrac{(\sqrt{3}/2)La(Lc-Lb)}{(K_2/1-k)-K_3}\right), 0\right) & (k \in (0,1) \text{ and } k \to 1). \end{cases} \tag{49}$$

In summary, the position estimation error and its range in each inductance region are calculated in this section, which approach zero when $k \to 1$. Therefore, we can conclude that the estimation performance is undoubtedly affected by inductance asymmetric reasons, but when the degree of asymmetric is low, the estimation error can be neglected and this will not severely affect the sensorless operation.

As discussed previously, only by measuring the full-cycle inductance of each phase, can the rotor position be estimated and the corresponding sensorless control scheme be implemented. Considering the phase current slope difference-based full-cycle inductance estimation method, the phase inductance at standstill, with rotating shaft conditions, and normal driving operation mode can all be real time measured for rotor position estimation. Therefore, this method to some extent, possesses good potential for extending the sensorless operation range for SRM drives. That is, on one hand, this method can estimate the initial rotor position at standstill and with rotating shaft conditions accurately and start up the SRM without starting hesitation; on the other hand, this sensorless method can drive the SRM from standstill to high-speed operation and the rotor position can be real time estimated. In order to verify our sensorless method, the corresponding experimental results are presented and analyzed in the following section.

IV. Experimental Verification

A. Experimental Implementation

The proposed sensorless scheme has been successfully implemented on an experimental SRM driving system. The overall system configuration of the proposed sensorless drive is shown in Fig. 14 and the hardware arrangement of the system is shown in Fig. 15.

1) Hardware Implementation: As shown in Fig. 15, the hardware system consists of the following units.

1) A three-phase 12/8 structured SRM is used.

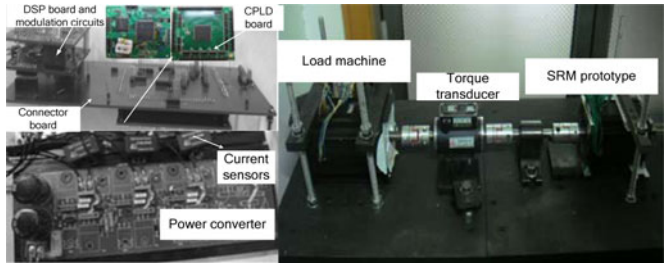


Fig. 15. Hardware of the experimental system.

2) The load machine is a switched reluctance generator (SRG). But in experiment, the SRG is not controlled, which is running synchronously with the SRM drive. The load torque is mainly produced by the axial friction force of the two machines and the other mechanical coupling parts.
3) The DSP2812 and CPLD hybrid controller is used for programming the sensorless method and executing control algorithm.
4) A three-phase asymmetrical half bridge with its drive circuits is used as the power converter.
5) Three current sensors and current modulation circuits are used for measuring the three phase current.
6) The position sensors are used for providing real position reference.

2) Software Implementation: As shown in Fig. 14, the software within the dashed frame is implemented by DSP and complex programmable logic device (CPLD) hybrid controller, which consists of the following parts.

1) The full-cycle inductance identification algorithm is used for estimating three-phase inductance.
2) The phase inductance subregion method and the inductance vector-based method are used for rotor position estimation.
3) The rotational speed is calculated from the estimated position angle.
4) Turn-ON angle $\theta_{\rm on}$ and turn-OFF angle $\theta_{\rm off}$ are chosen according to the speed for angle position control.
5) A PI controller is designed for providing the chopping current reference, and through CCC, the speed closed-loop can be implemented.

3) The Experimental Conditions: As discussed in Section II, the proposed method neglects the magnetic saturation of SRM; thus, the method is more suitable for no-load and light-load application. To verify the proposed method, several experiments are implemented under the conditions as shown in Table III.

B. Experimental Results Under Different Operation Conditions

Based on the experimental system, related experiments have been done under different operation conditions. In the following discussion, if not otherwise specified, the rotor position is in mechanical degrees relative to the unaligned positions of phase-A.

1) Rotor Position Estimation at Standstill: The application performance of the method for initial rotor position estimation is

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TABLE III
EXPERIMENTAL CONDITIONS

SRM Prototype	A rated 3kW / 12/8 structure SRM
Applied DC bus voltage	20 V
Load condition	Light load (about 5% of full load)
Injection frequency	3.33kHz
current sampling interval	0.00002048s
The chopping interrupt cycle of DSP Timer	100us
Operation speed range	0- 2000rpm

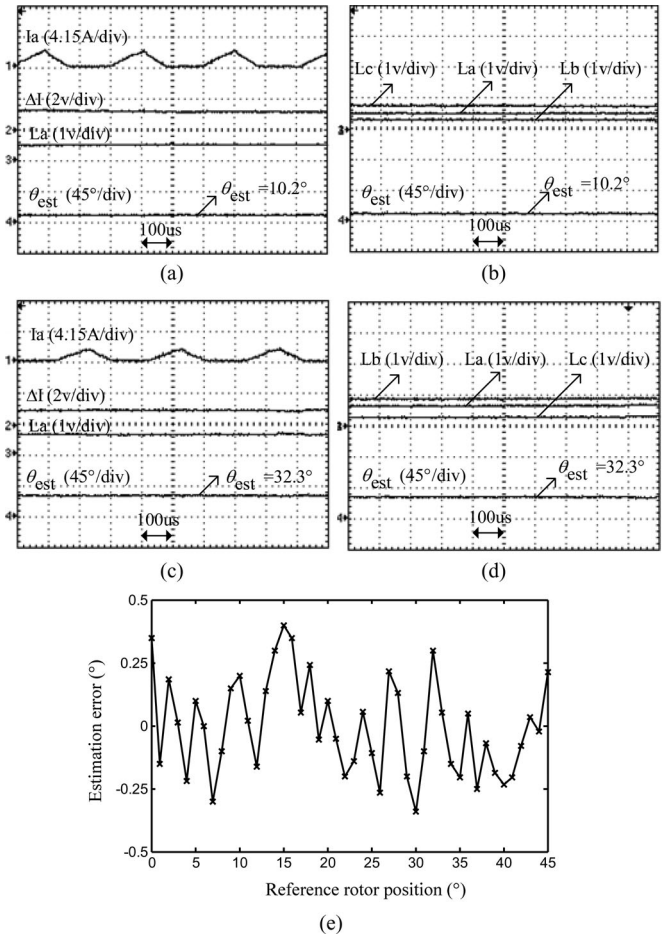


Fig. 16. Experimental results at standstill. (e) Rotor position estimation error at standstill.

first verified. At first, exciting phase-A for a short while and let the rotor rotate to the aligned position of phase-A. Then, according to this benchmark, the rotor can be placed in any reference initial position. Fig. 16 presents the testing results at different initial rotor positions. The actual reference position is at 10° in Fig. 16(a) and (b) and 32° in Fig. 16(c) and (d), respectively. As shown in Fig. 16(a) and (c), all phases are injected with voltage pulses of 3.33-kHz frequency; then, the phase current pulses are measured and the filtered current slope difference are obtained for inductance and rotor position estimation. The estimation results of the three-phase inductance and the rotor position are shown in Fig. 16(b) and (d), where at the given reference rotor

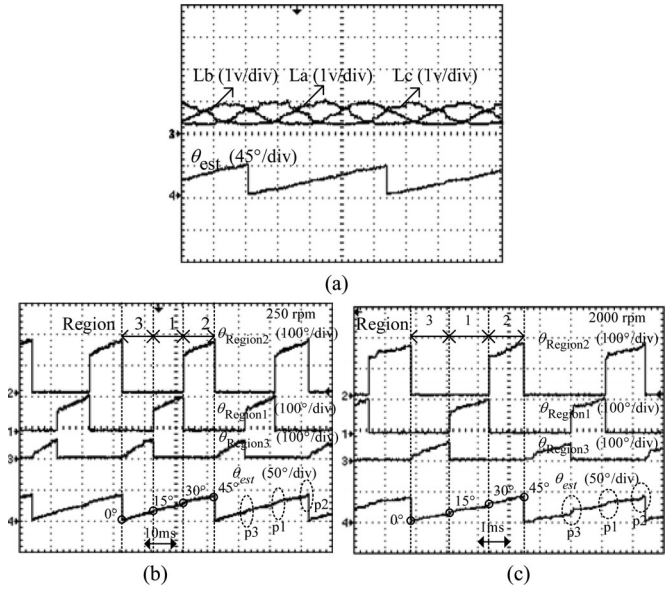


Fig. 17. (a) Estimated three-phase inductance. (b) Position estimation process at 250 r/min. (c) Position estimation process at 2000 r/min.

position, the phase inductance of each phase is constant and the corresponding rotor position can be calculated.

In this way, at standstill, the rotor position can be estimated at any given initial rotor position. In experiment, rotor positions in a complete mechanic cycle with steps of one degree are detected. Fig. 16(e) illustrates the estimation error as compared to the actual positions, which shows the estimation error is less than 0.5°.

2) Rotor Position Estimation at Running Conditions: As discussed, the running condition consists of normal driving operation mode and rotating shaft conditions mode. With regard to the rotor position estimation process, they are only different in inductance identification stage. Based on the inductance identification method, the estimated three-phase inductance profiles are shown in Fig. 17(a). As can be seen, the three-phase inductance is almost symmetric. Thus, in this paper, the estimation error caused by inductance asymmetric can be neglected and it will not severely affect the sensorless operation. Fig. 17(b) and (c) illustrates the position estimation process as represented in (30) at 250 and 2000 r/min, respectively. As can be seen from these figures, the rotor position is estimated in different inductance regions, respectively. And the full-cycle rotor position can be constructed by combining the estimated position values in each region.

a) Rotor Position Estimation at Rotating Shaft Conditions Mode: To verify the effectiveness of the position estimation algorithm at rotating shaft conditions, Fig. 18 shows the position estimation results at 300, 1000, and 2000 r/min, respectively. As shown in Fig. 18(a), (c) and (e), the process of the full-cycle inductance identification and rotor position estimation at different rotor speeds are presented. From the top to the bottom, the traces show the phase current pulses, filtered current slope difference, phase inductance, and the estimated rotor position. As can be seen, in this operation mode, high-frequency voltage

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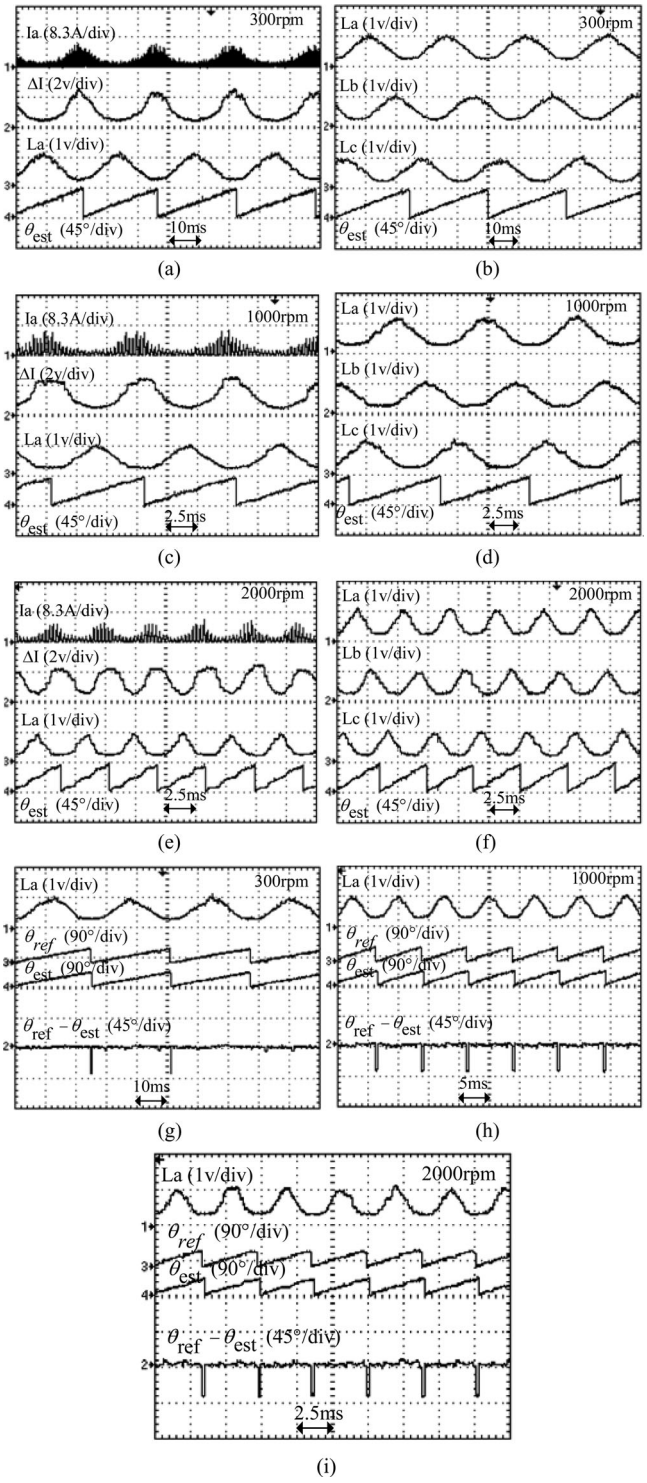


Fig. 18. Experimental results at rotating shaft conditions. (a), (c), and (e) Inductance identification process at 300, 1000, and 2000 r/min, respectively. (b), (d), and (f) Identified three-phase inductance and the estimated rotor position at 300, 1000, and 2000 r/min, respectively. (g)–(i) Comparison between the actual position, estimated position, and their errors at 300, 1000, and 2000 r/min, respectively.

pulses were injected in phase coils continuously for inductance identification. Fig. 18(b), (d) and (f) shows the estimation results of the three-phase inductance and rotor position at different rotor speeds, respectively. As shown in these figures, the three-phase inductance at different speeds is estimated accurately and is almost symmetric. In order to evaluate the accuracy of the position estimation method at this operation condition, Fig. 18(g)–(i) shows the comparison between the actual position, estimated position, and their errors at different rotor speed, respectively. As can be seen in the figures, both actual position and estimated position are rounded between 0° and 45° mechanical degrees for each cycle, with the increasing of rotor speed, the number of detecting pulses decrease, and deteriorate the estimation accuracy of the phase inductance, which makes the estimated phase inductance and rotor position more discrete. The experimental results show that the proposed estimation method possesses good accuracy. However, it is more accurate at low speed rather than high speed.

b) Rotor Position Estimation at Normal Driving Operation Mode: Fig. 19 shows the details of rotor position estimation performance under sensorless closed-loop control mode at different reference speed including 300, 1000, and 2000 r/min, respectively. Fig. 19(a), (c), and (e) shows the phase inductance identification process and the estimated position at each reference speed, respectively. As can be seen from their first trace, in nonenergized regions, high-frequency voltage pulses are injected, while in energized regions, the phase current is regulated by the chopping current provided by speed closed-loop PI controller. Under this control mode, the phase inductance can be identified by using the current slope difference method; the identified three-phase inductances at different speed are shown in Fig. 19(b), (d), and (f), respectively. Furthermore, in Fig. 19(g)–(i), the estimated rotor position is compared with the accrual rotor positions. It can be noted that the rotor position is accurately detected at different given speeds. Here, the reference rotor position is displayed only for comparison, and the system is completely working under sensorless control.

As compared with Fig. 18(g)–(i), the position estimation results under normal driving operation mode seems similar with that at rotating shaft conditions. As shown in Fig. 19(g)–(i), it is clear that the estimation performance is better at lower speed range. That is because: 1) the lower the speed, the more the detecting pulses injected at nonenergized region and the more the numbers of chopping intervals at energized regions, which makes the estimated phase inductance and rotor position more continuous and accurate; and 2) as can be seen from Fig. 17(b) and (c), the estimation error mainly occurs around the phase inductance junction points p1, p2, and p3; with the increasing of speed, the number of sample points of phase inductance decreases, which may result in larger irregular phase shift of the junction points; consequently, the estimation accuracy will reduce. This problem also exits in the cases under rotating shaft conditions. Though the position estimation error exists in both operation modes, the position estimation is accurate enough to control the machine. The dynamic performance of sensorless operation is discussed in the following section.

C. Sensorless Operation Performance Verification

To verify the dynamic performance of the proposed sensorless method, experiments of sensorless closed-loop control are implemented on the experimental system.

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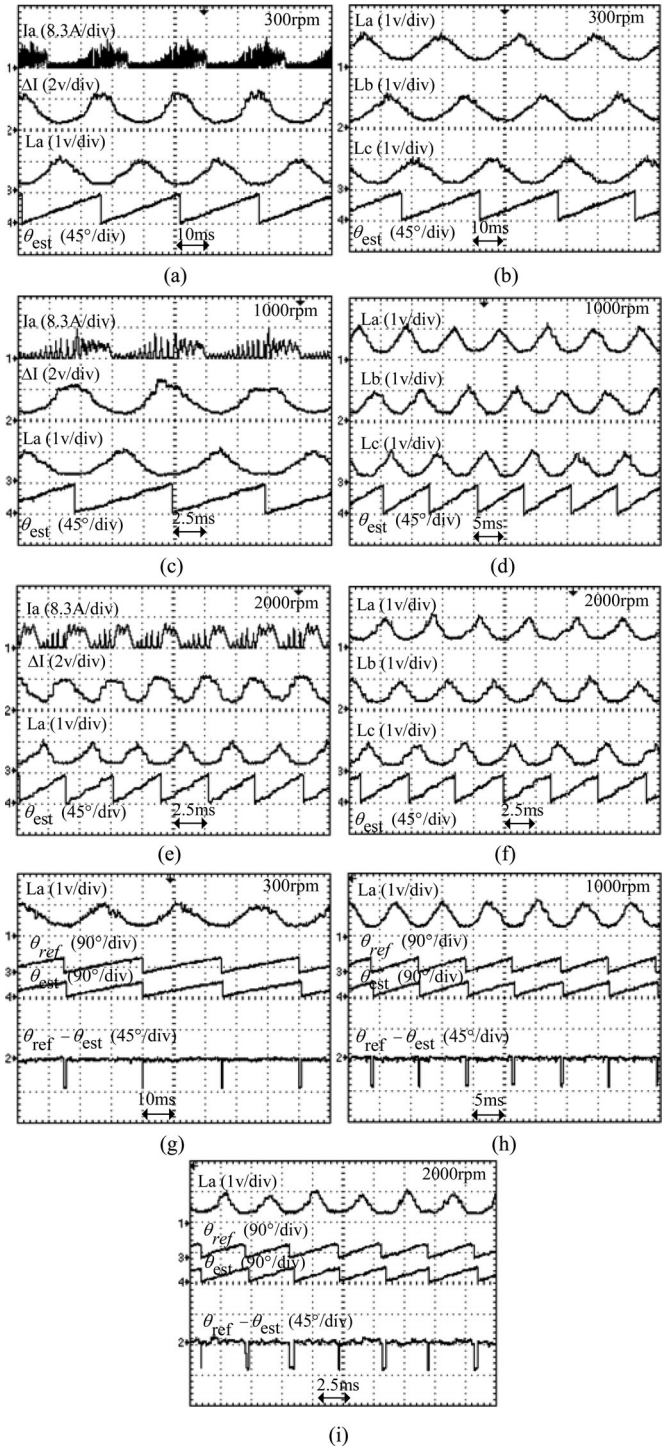


Fig. 19. Experimental results at normal driving operation mode. (a), (c), and (e) Inductance identification process at 300, 1000, and 2000 r/min, respectively. (b), (d), and (f) Identified three-phase inductance at 300, 1000, and 2000 r/min, respectively. (g)–(i) Comparison between the actual position, estimated position, and their errors at 300, 1000, and 2000 r/min, respectively.

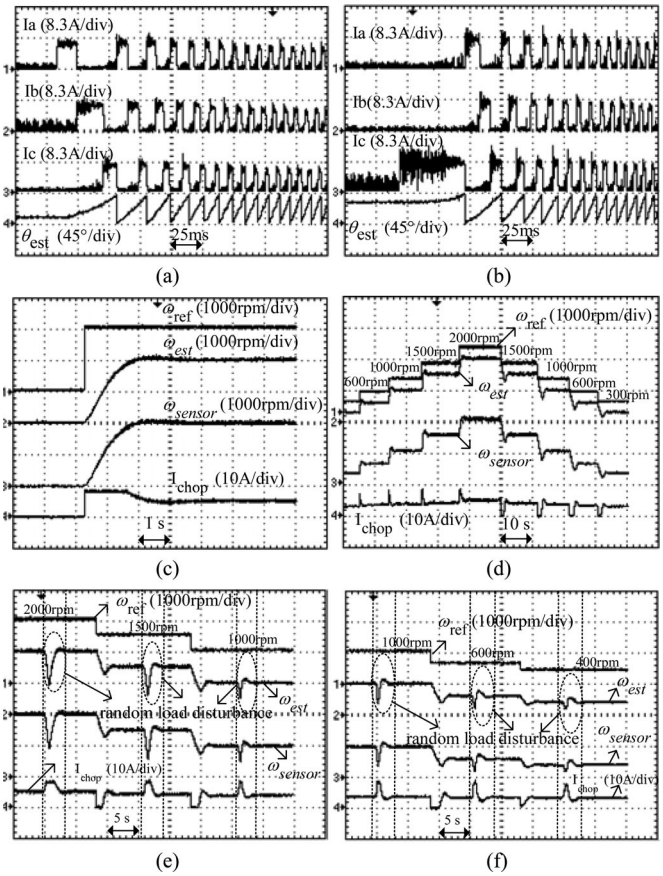


Fig. 20. Experimental results of sensorless operation. (a) and (b) Sensorless starting process at different initial rotor positions, 10° and 32°, respectively. (c) Rotor speed waveform for sensorless starting up. (d) Measured sensorless speed transients. (e) and (f) Transient response to random load disturbance at 2000, 1500, 1000, 600, and 400 r/min, respectively.

Fig. 20(a) and (b) shows the testing results of the sensorless starting process at different initial rotor positions, 10° and 32°, respectively. In these figures, the three-phase current and the estimated rotor positions are shown. The rotor speed waveform for sensorless starting is shown in Fig. 20(c). The experimental results verify the corresponding initial rotor position estimation performance, as described in Fig. 16, and confirm that the estimation method can start the SRM at any initial rotor position without any starting hesitation.

Fig. 20(d) depicts the speed transient performance of the sensorless closed-loop control method. As can be seen, the sensorless scheme works well at a series of speed transients both in acceleration and deceleration modes. Clearly, the sensorless control performance is comparable with those of position sensors control mode.

Finally, Fig. 20(e) and (f) illustrates the sensorless drive's response to steps of random load disturbance at 2000, 1500, 1000, 600, and 400 r/min, respectively, which shows that the sensorless drive system possesses good capacity of resisting load disturbance.

V. Conclusion

A new sensorless control method for SRM drives has been presented in this paper. The three-phase full cycle inductance are identified by phase current slope deference method, which can be used for determining the suitable rotor position estimation regions based on the proposed phase inductance subregion method and estimating the rotor position by orthogonal decomposition of the phase inductance composed vector. This

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sensorless method eliminates the problem of starting hesitation and offers adequate accuracy in rotor position estimation even in slightly inductance asymmetric conditions, which can drive the SRM from standstill to high-speed operation smoothly. Besides, by utilizing the speed closed-loop technique, the sensorless method works well at a series of speed transients both in acceleration and deceleration and possesses good capacity of resisting load disturbance. In addition, this sensorless method requires no additional hardware or complex computation or *a priori* knowledge on magnetic characteristics of SRM, so it is relatively easy to implement and can be easily modified to fit SRMs with different poles configurations. These features make the proposed method very practical, reliable, and cost effective and could be acceptable in many low-cost variable-speed applications. Experimental results fully verify the proposed sensorless scheme and demonstrate its advantages.

APPENDIX

$$K_1 = Lc^2 + Lb^2 - LcLb + La'La - \frac{(La + La')(Lb + Lc)}{2} \tag{A1}$$

$$K_2 = Lc^2 + Lb^2 + La^2 - LcLb - LaLb - LaLc \tag{A2}$$

$$K_3 = \left(La^2 - \frac{1}{2}La(Lb + Lc) \right) \tag{A3}$$

$$K_4 = \left[La' - \frac{1}{2}(Lb + Lc) \right] \left[La - \frac{1}{2}(Lb + Lc) \right] \tag{A4}$$

$$K_5 = -La \left[La - \frac{1}{2}(Lb + Lc) \right] \tag{A5}$$

$$K_6 = \left[La - \frac{1}{2}(Lb + Lc) \right]^2 + \frac{3}{4}(Lb - Lc)^2. \tag{A6}$$

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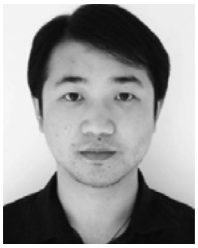
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